Optimal Energy and Reserve Scheduling of Pumped-Storage Power Plants Considering Hydraulic Short-Circuit Operation

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***Abstract*—This paper presents a mixed-integer model for the hourly energy and reserve scheduling of a price-taker and closed-loop pumped-storage hydropower plant operating in hydraulic short-circuit mode. The plant participates in the spot market and in the secondary regulation reserve market, taking into account the regulation energy due to the real-time use of the regulation-up and -down reserves. The proposed model is used to compare the maximum theoretical income of the plant with and without considering hydraulic short-circuit operation. Numerical results demonstrate that the operation in hydraulic short-circuit mode could help significantly to enlarge the income of the power plant and that the secondary regulation reserve market might be the main source of revenue in a realistic setting characterized by a high level of renewable energy sources in the generation mix.**

***Index Terms*—Hydraulic short-circuit technology, pumped storage plant, secondary regulation service.**

NOMENCLATURE

Subscripts/Superscripts:

c	subscript for hydropower unit, running from 1 to C.
d	superscript that indicates that the magnitude is related to generation/discharge.
k	subscript for discrete generation curves.
p	superscript that indicates that the magnitude is related to consumption/pumping.
sec	superscript that indicates that the magnitude is related to the secondary regulation service (reserve or energy).
t	subscript for hourly period, running from 1 to T.

Parameters:

cSU_c^d	start-up cost of the turbine of hydro unit c, €.
cSU_c^p	start-up cost of the pump of hydro unit c, €.
$\delta_{c,k}^d$	energy coefficient in generating mode, MW/hm^3/h.
fv	target water volume in last hour of time horizon, hm^3.
$\overline{g}_{c,k}^d$, $\underline{g}_{c,k}^d$	maximum and minimum technical power generation, MW.
$\overline{g}_{c,k}^p$, $\underline{g}_{c,k}^p$	maximum and minimum technical power consumption, MW.
l_t	time length of period t, 1 h.
$\lambda_{D,t}$	day-ahead electricity market price, €/MW·h.
$\lambda_{S,t}$	secondary regulation reserve market price, €/MW.
$\lambda_{\mathrm{up},t}$	secondary regulation-up energy market price, €/MW·h.
$\lambda_{\mathrm{dw},t}$	secondary regulation-down energy market price, €/MW·h.
M	a big number, 10^6.
$\overline{\eta}_{c,k}^d$, $\underline{\eta}_{c,k}^d$	turbine efficiency at maximum and minimum flow, %.
$\eta_{c,k}^p$	pump efficiency, %.
$\overline{q}_{c,k}^d$, $\underline{q}_{c,k}^d$	maximum and minimum technical water discharge, hm^3/h.
$\overline{q}_{c,k}^p$, $\underline{q}_{c,k}^p$	maximum and minimum technical pumped water, hm^3/h.
ρ_t^{up}, ρ_t^{dw}	percentage of real-time use of secondary regulation-up and down reserves.
R_t^{SM}	ratio between secondary regulation-up reserve and total regulation reserve, set by the transmission system operator (TSO) in advance.
$\overline{v}$, $\underline{v}$	maximum and minimum technical water storage limits of upper reservoir due to design characteristics, hm^3.

Positive Variables:

$g_{c,t}^d$	power generation, MW.
$g_{c,t}^p$	power consumption, MW.
$q_{c,t}^d$	total water discharge, which includes water flow for real-time use of secondary regulation-up and down reserves, hm^3/h.
$q_{c,t}^p$	pumped water, hm^3/h.
$qs_{c,t}^d$	water discharge above the minimum technical limit, hm^3/h.
$g_{c,t}^{\mathrm{sec,up}}$	secondary regulation-up reserve, MW.
$g_{c,t}^{\mathrm{sec,dw}}$	secondary regulation-down reserve, MW.
$aux_g_{c,t}^{\mathrm{sec}}$	auxiliary variable associated with $g_{c,t}^{\mathrm{sec,dw}}$, MW.

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$e_t^{\mathrm{sec,up}}$ secondary regulation-up reserve requested in real time by the TSO, MW·h.
$e_t^{\mathrm{sec,dw}}$ secondary regulation-down reserve requested in real time by the TSO, MW·h.
v_t water volume of the reservoir, hm^3.

Binary Variables:

d_t used for the discretization of the hydropower generation/consumption curves.
$u_{c,t}^d$ On/Off state in generating mode.
$u_{c,t}^p$ On/Off state in pumping mode.
$y_{c,t}^d$ 1 if the turbine of a hydro unit is started-up, 0 otherwise.
$y_{c,t}^p$ 1 if the pump of a hydro unit is started-up, 0 otherwise.
ϕ_t 1 if there is more secondary regulation-up reserve requested in real time than regulation-down reserve, 0 otherwise.

I. Introduction

PUMPED storage hydropower plants (PSHPs) are deemed as a mature economic and large-scale feasible solution to store energy in an efficient way comparing to other technologies such as supercapacitors or electrochemical batteries [1]. Typical ratings of PSHPs are around 1000 MW and, in most cases, they are capable to discharge water at rated power during several hours [2]. In addition to this, they can participate in ancillary services markets, offering regulating frequency reserve, emergency reserves, etc.

However, the construction of a PSHP requires high investment costs (between 470 to 2170€/kW [3]), which could be a barrier for commissioning new projects. Many factors could jeopardise the recovery of investment costs [4]: 1) income from price arbitrage between peak and off-peak hours is enough in just a few electricity markets [3], 2) PSHPs are obliged to pay grid fees for the usage of transmission lines when they are consuming, as it occurs in many power systems such as the Spanish system or 3) high water fees for utilising rivers or lakes reduce profits of PSHP operation.

Several researches have shown that PSHPs can hardly recover their capital costs from the price arbitrage or load-shifting traditional operation (see [3] in deregulated electricity markets or [5] in a centralised electricity market). In the context of liberalised electricity systems, the participation of PSHPs in other markets, mainly in the provision of regulation service, becomes critical to improve the profitability of the pumped-storage facility [6], [7]. In [8], secondary regulation reserve market (SM) proves to be the most important income source for a PSHP in a real day of the Portuguese electricity system and in other two synthetic 24-h scenarios.

Currently, most PSHPs in the world (hereafter referred to as conventional PSHPs) pump water at a fixed speed and therefore, are not able to provide load-frequency control in pumping mode. However, in order to help the integration of more intermittent renewable energy sources, during last years the hydropower industry has put special attention in technical solutions which allow providing load-frequency control while pumping, such as variable speed [9] and hydraulic short-circuit (SC) operation [10].

The aim of this paper is twofold: 1) to present a mixed integer linear programming (MILP) model for the energy and reserve scheduling of a realistic closed-loop[1] and daily-cycle PSHP, considering hydraulic SC operation, 2) to show and to quantify the extent to which hydraulic SC operation is able to increase the income of a price-taker PSHP that participates both in the day-ahead energy market (DM) also called spot market, and the secondary regulation service of the Spanish electricity power system. The secondary regulation service in the Spanish electricity market comprises two different concepts: i) a reserve market, which takes place at 5:30 pm of the previous day (after having cleared the spot market) where power reserve availability is remunerated by the marginal market price, and ii) power reserve delivery in real time, which is remunerated by the marginal price of the tertiary regulation market [11]. The model, used to estimate the maximum theoretical income in the above-mentioned markets and services, is based on the one proposed in [12], which has been revised in order to consider several units within the same PSHP, and the head dependency of both power and available spinning reserve, as well as to allow for the operation in hydraulic SC mode instead of variable speed operation.

Diverse optimization models for bidding and/or scheduling conventional PSHPs in the day-ahead energy market [3], [13]–[15] and simultaneously in the energy and reserve markets [16]–[21] have been published in the literature. However, to the authors' knowledge, there are no papers where the optimal scheduling of PSHPs operating in hydraulic SC mode and participating in the energy and reserve markets has been explicitly formulated [7].

Main contributions with respect previous studies are the following: 1) an MILP model for the energy and reserve scheduling of a PSHP considering hydraulic SC operation and the real-time use of reserves is proposed in the paper. The model considers a variable number of PSHP units, the units' start-up costs and the head dependency in both power and available spinning reserve, and 2) the increase in the maximum theoretical income of a PSHP due to the operation in hydraulic SC mode has been quantified. The features considered in the proposed model have not been taken into account in previous studies. Therefore, the conclusions derived from the presented results can help to assess the profitability of the different configurations of PSHPs.

The paper is organised as follows: Section II describes the operation of PSHPs in hydraulic SC. Section III presents the scheduling model formulation. The application of the model to a realistic PSHP is presented in Section IV and finally, conclusions are described in Section V.

II. Pumped-Storage Power Plants in Hydraulic SC Operation

To the best of our knowledge, Kops II, Malta, Geesthacht and Luenersee are the only four PSHPs that are operated in

[1]Closed-loop PSHPs do not receive natural water inflows in the upper reservoir and do not release water from the lower reservoir.

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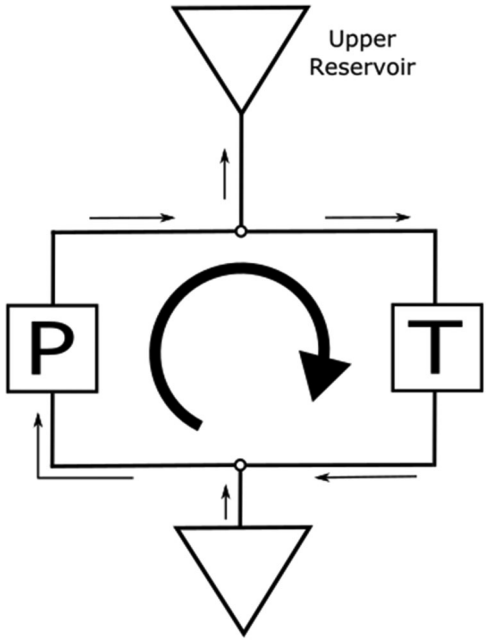


Fig. 1. Pumped-storage power plant operating in hydraulic SC mode.

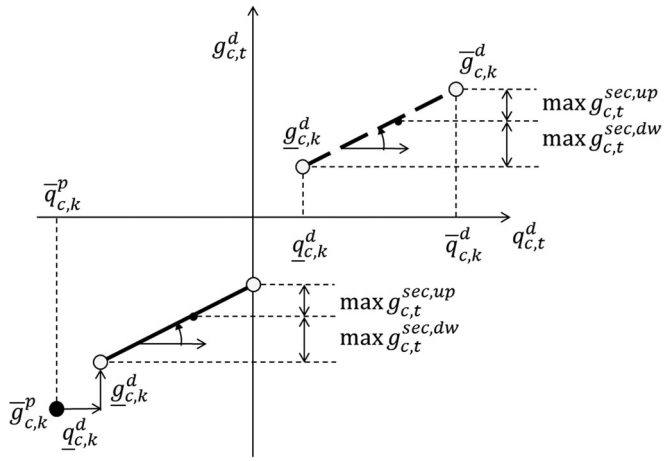


Fig. 2. I/O curve of a ternary unit in hydraulic SC. Dashed line for generating mode, black dot for pumping mode and solid line for SC mode.

TABLE I
MODES OF OPERATION OF DIFFERENT PSHPS

PSHP	Pumping	Generating	Hydraulic SC
Conv	fixed speed	Francis pump-turbine	n/a
SC-B	fixed speed	Francis pump-turbine	two or more units
SC-T	fixed speed	Pelton	one or more units

hydraulic SC mode in the world. Kops II, owned by Vorarlberger Illwerke (Austria), was commissioned in 2008 and it is the most documented [22], [23]. The power station scheme of a PSHP operating in hydraulic SC mode with a single ternary unit is presented in Fig. 1. A ternary unit is composed by a Pelton turbine, a fixed-speed pump and a motor/generator electric machine, all connected vertically. Thanks to a clutch, both the turbine and the pump of the ternary unit can be operated simultaneously. The main advantage of this is that the PSHP is able to regulate power (and therefore, to participate in the secondary regulation service) while it is consuming energy, with a power regulation range equal to that of the turbines in operation. In other words, the PSHP operating in hydraulic SC mode can be considered as a controllable load, contributing to a much better integration of intermittent renewable energy and to balance generation and demand in real time better than conventional PSHPs.

From the point of view of modelling, three operating modes must be taken into account: generating, pumping and hydraulic SC modes. The input/output curve (I/O curve) of a single ternary unit, for a certain water level in the upper reservoir is presented in Fig. 2. In generating mode (dashed line), water is discharged from the upper to the lower reservoir through the Pelton turbine (while the pump is off) and energy is injected to the grid. When the unit is operated in pumping mode (black dot), water is pumped from the lower to the upper reservoir through the fixed-speed pump and energy is consumed from the grid. Finally, when the unit is operated in hydraulic SC mode (solid line, obtained by means of a simple translation of the generating mode curve in such a way that origin is moved to the point that corresponds to the pumping mode at fixed rate), both turbine and pump are on. A percentage of water that is taken from the lower reservoir is pumped to the upper reservoir and the rest is passed through the turbine, delivering energy to the grid and regulating the output power if necessary. The maximum secondary regulation-up, $\max\ g^{\mathrm{sec,up}}_{c,t}$ and down, $\max\ g^{\mathrm{sec,dw}}_{c,t}$ reserves, in generating and hydraulic SC modes are shown in Fig. 2 (it is not possible to regulate in pumping mode since the pump operates with fixed speed). Reserves that are requested in real time are supplied as following: regulation-up (resp. down) energy is delivered by increasing (resp. reducing) the water discharge through the turbines.

If the PSHP is composed by several ternary units, all pumps and turbines can be operated simultaneously in hydraulic SC mode, with a total regulation range equal to that of all turbines in operation. Finally, although as far as we know there is no PSHP operating in hydraulic SC mode with binary units (SC-B) (a binary unit is composed by a reversible Francis pump-turbine and a motor/generator electric machine), theoretically, they could be also operated in the said mode, as long as they comprise two or more units (at least one unit must operate in pumping mode and another in generating mode). Nonetheless, the power regulation range of PSHPs operating in hydraulic SC-B is significantly lower than with ternary units, since each binary unit must be operated as a turbine or as a pump but not simultaneously. Furthermore, the power regulation range of a Francis pump-turbine is notably lower than that of a Pelton turbine. In Section IV, the operation and income of a PSHP with ternary units operating in hydraulic SC mode are compared with those of a conventional PSHP with binary units and with a PSHP operating in hydraulic SC-B.

Table I summarises the main features of the pumping, generating and hydraulic SC operating modes of the configurations

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of conventional PSHPs (Conv), PSHPs operating in hydraulic SC-B and PSHPs operating in hydraulic SC mode with ternary units (SC-T) considered in Section IV.

III. Model Formulation

In this section, three models are presented. Section III-A shows the model formulation of PSHPs operating in hydraulic SC-T (SC-T PSHP) whereas Section III-B describes the model formulation of PSHPs operating in hydraulic SC-B (SC-B PSHP). Differences between a PSHP operating in hydraulic SC-T and SC-B are explained in Section II. Finally, Section III-C presents the model formulation of conventional PSHPs (Conv PSHP).

A. Model Formulation for SC-T PSHP

1) Objective Function: The objective function consists in maximizing the net income of a PSHP participating in the DM and the secondary regulation service in the framework of the Spanish electricity market [11]. In (1), the first line expresses the net income for the regulation energy due to the real-time use of the secondary regulation reserves. The second line expresses the net income in the DM and the income for reserves put at disposal of the TSO in the SM. And, finally, the third line expresses the start-up costs of turbines and pumps, respectively,

$$\begin{aligned}\text{Max}\quad z = & \sum_t \Big\{ \Big(e_t^{\text{sec,up}} \cdot \lambda_{\text{up},t} - e_t^{\text{sec,dw}} \cdot \lambda_{\text{dw},t} \Big) \\ & + \sum_c \Big[\big(g_{c,t}^d - g_{c,t}^p \big) \cdot \lambda_{D,t} + \Big(g_{c,t}^{\text{sec,up}} + g_{c,t}^{\text{sec,dw}} \Big) \\ & \cdot \lambda_{S,t} - cSU_c^d \cdot y_{c,t}^d - cSU_c^p \cdot y_{c,t}^p \Big] \Big\}. \end{aligned} \tag{1}$$

2) Water Balance: The water balance, (2) does not include any water inflow because a closed-loop PSHP is considered in this article. Furthermore, in order to express all terms of the water balance equation with the same dimension, time duration of each period t is included with flow variables even though its value is one hour. In addition to this, water limits are imposed to water volume in the upper reservoir, v_t in all time period t, (3). Water balance and limits in the lower reservoir are not included in the model formulation for the shake of clarity,

$$v_t = v_{t-1} + \sum_c l_t \cdot \big(q_{c,t}^p - q_{c,t}^d \big) \quad \forall t \tag{2}$$

$$\underline{v} \le v_t \le \overline{v} \quad \forall t. \tag{3}$$

In the upper reservoir, a target water volume is imposed, (4), where T stands for the last period of the time horizon. The target water volume is deemed as input data to the model and equals initial volume in Section IV

$$v_t = fv \quad \forall t = \text{T}. \tag{4}$$

3) Hydropower Generation: Generation/consumption curves of each unit of SC-T PSHPs are modelled by equations (7)–(24) (see Fig. 2 in Section II). The presented formulation for generation/consumption curves follows the one proposed in [24].

Two generation/consumption curves, k_1 and k_2 have been considered in the formulation, and correspond to two different reservoir levels. Each curve is selected using the binary variable, d_t that depends on the water volume of the upper reservoir, (5) and (6). The power generation is calculated from (7)–(10), as a function of d_t and $u_{c,t}^d$. With equations (11) and (12), the maximum and minimum water discharges, which depend on the water level in the upper reservoir, are considered. In addition to this, the water discharge takes into account not only the flow that is used to produce the energy committed or scheduled in the spot market but also the extra flow that is required for providing regulation-up energy, and the flow that is kept upstream in order to provide regulation-down energy, (13)–(16). The water discharge used for deploying the secondary regulation energy also depends on the water level in the upper reservoir. This has been modelled by using the appropriate energy coefficient of each generation curve, $\delta_{k_1}^d$ and $\delta_{k_2}^d$. Head dependency in consumed power and pumped water is included in (17)–(24). Start-up decisions are modelled in (25) for turbines and (26) for pumps. In the case that the maximum variation of the water level in the upper reservoir can be negligible in comparison with the available gross head, the corresponding equations can be derived in a straightforward manner from those used in this paper

$$v_t \ge \underline{v} \cdot (1 - d_t) + \left(\underline{v} + \frac{\overline{v} - \underline{v}}{2} \right) \cdot d_t \quad \forall t \tag{5}$$

$$v_t \le \overline{v} \cdot d_t + \left(\underline{v} + \frac{\overline{v} - \underline{v}}{2} \right) \cdot (1 - d_t) \quad \forall t \tag{6}$$

$$g_{c,t}^d \le u_{c,t}^d \cdot \underline{g}_{c,k}^d + qs_{c,t}^d \cdot \delta_k^d + M \cdot d_t \quad \forall c, t, k = k_1 \tag{7}$$

$$g_{c,t}^d \ge u_{c,t}^d \cdot \underline{g}_{c,k}^d + qs_{c,t}^d \cdot \delta_k^d - M \cdot d_t \quad \forall c, t, k = k_1 \tag{8}$$

$$g_{c,t}^d \le u_{c,t}^d \cdot \underline{g}_{c,k}^d + qs_{c,t}^d \cdot \delta_k^d + M \cdot (1 - d_t) \quad \forall c, t, k = k_2 \tag{9}$$

$$g_{c,t}^d \ge u_{c,t}^d \cdot \underline{g}_{c,k}^d + qs_{c,t}^d \cdot \delta_k^d - M \cdot (1 - d_t) \quad \forall c, t, k = k_2 \tag{10}$$

$$qs_{c,t}^d \le u_{c,t}^d \cdot M \quad \forall c, t \tag{11}$$

$$qs_{c,t}^d \le \big(\overline{q}_{c,k_1}^d - \underline{q}_{c,k_1}^d \big) \cdot (1 - d_t) + \big(\overline{q}_{c,k_2}^d - \underline{q}_{c,k_2}^d \big) \cdot d_t \quad \forall c, t \tag{12}$$

$$q_{c,t}^d \le u_{c,t}^d \cdot \underline{q}_{c,k}^d + qs_{c,t}^d + M \cdot d_t + \frac{1}{\delta_k^d} \cdot \big(\rho_t^{\text{up}} \cdot g_{c,t}^{\text{sec,up}} - \rho_t^{\text{dw}} \cdot g_{c,t}^{\text{sec,dw}} \Big) \quad \forall c, t, k = k_1 \tag{13}$$

$$q_{c,t}^d \ge u_{c,t}^d \cdot \underline{q}_{c,k}^d + qs_{c,t}^p - M \cdot d_t + \frac{1}{\delta_k^d} \cdot \big(\rho_t^{\text{up}} \cdot g_{c,t}^{\text{sec,up}} - \rho_t^{\text{dw}} \cdot g_{c,t}^{\text{sec,dw}} \Big) \quad \forall c, t, k = k_1 \tag{14}$$

$$q_{c,t}^d \le u_{c,t}^d \cdot \underline{q}_{c,k}^d + qs_{c,t}^d + M \cdot (1 - d_t) + \frac{1}{\delta_k^d} \cdot \big(\rho_t^{\text{up}} \cdot g_{c,t}^{\text{sec,up}} - \rho_t^{\text{dw}} \cdot g_{c,t}^{\text{sec,dw}} \Big) \quad \forall c, t, k = k_2 \tag{15}$$

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$$q^d_{c,t} \geq u^d_{c,t} \cdot \underline{q}^d_{c,k} + qs^d_{c,t} - M \cdot (1 - d_t) + \frac{1}{\delta^d_k} \cdot (\rho^{\mathrm{up}}_t \cdot g^{\mathrm{sec,up}}_{c,t} - \rho^{\mathrm{dw}}_t \cdot g^{\mathrm{sec,dw}}_{c,t}) \quad \forall c, t, k = k_2 \tag{16}$$

$$g^p_{c,t} \leq u^p_{c,t} \cdot \overline{g}^p_{c,k} + M \cdot d_t \quad \forall c, t, k = k_1 \tag{17}$$

$$g^p_{c,t} \geq u^p_{c,t} \cdot \overline{g}^p_{c,k} - M \cdot d_t \quad \forall c, t, k = k_1 \tag{18}$$

$$g^p_{c,t} \leq u^p_{c,t} \cdot \overline{g}^p_{c,k} + M \cdot (1 - d_t) \quad \forall c, t, k = k_2 \tag{19}$$

$$g^p_{c,t} \geq u^p_{c,t} \cdot \overline{g}^p_{c,k} - M \cdot (1 - d_t) \quad \forall c, t, k = k_2 \tag{20}$$

$$q^p_{c,t} \leq u^p_{c,t} \cdot \overline{q}^p_{c,k} + M \cdot d_t \quad \forall c, t, k = k_1 \tag{21}$$

$$q^p_{c,t} \geq u^p_{c,t} \cdot \overline{q}^p_{c,k} - M \cdot d_t \quad \forall c, t, k = k_1 \tag{22}$$

$$q^p_{c,t} \leq u^p_{c,t} \cdot \overline{q}^p_{c,k} + M \cdot (1 - d_t) \quad \forall c, t, k = k_2 \tag{23}$$

$$q^p_{c,t} \geq u^p_{c,t} \cdot \overline{q}^p_{c,k} - M \cdot (1 - d_t) \quad \forall c, t, k = k_2 \tag{24}$$

$$y^d_{c,t} \geq u^d_{c,t} - u^d_{c,t-1} \quad \forall c, t \tag{25}$$

$$y^p_{c,t} \geq u^p_{c,t} - u^p_{c,t-1} \quad \forall c, t. \tag{26}$$

4) Secondary Regulation Service: The secondary regulation service (also called secondary control or load-frequency control service) in the Spanish electricity market comprises the hourly power reserve of each participant, that is committed in a day-ahead reserve market, and the reserve requested in real-time by the TSO. The secondary regulation reserve bids are modelled through (27)–(31). Note that the auxiliary variable $aux_g^{\mathrm{sec}}_{c,t}$ avoids non-linearities in the formulation of the secondary regulation-down reserve due to the head dependency in the available spinning reserve. When a unit is off in generating mode ($u^d_{c,t} = 0$), there is no available secondary regulation reserve by (29), and $aux_g^{\mathrm{sec}}_{c,t}$ will take a value greater than or equal to the expression in square brackets in (28) and lower than M by (30). In addition, when a unit is on in generating mode ($u^d_{c,t} = 1$), $aux_g^{\mathrm{sec}}_{c,t}$ equals zero through (30). Moreover, the Spanish TSO requires the market agents to fulfil an hourly ratio between regulation-up reserve and total reserve, (31). When a bid is assigned in this market, the PSHP must be ready to use it

$$g^{\mathrm{sec,up}}_{c,t} \leq \left[\overline{g}^d_{c,k_2} \cdot d_t + \overline{g}^d_{c,k_1} (1 - d_t)\right] - g^d_{c,t} \quad \forall c, t \tag{27}$$

$$g^{\mathrm{sec,dw}}_{c,t} - aux_g^{\mathrm{sec}}_{c,t} \leq g^d_{c,t} - \left[\underline{g}^d_{c,k_2} \cdot d_t + \underline{g}^d_{c,k_1} (1 - d_t)\right] \quad \forall c, t \tag{28}$$

$$g^{\mathrm{sec,up}}_{c,t} + g^{\mathrm{sec,dw}}_{c,t} \leq M \cdot u^d_{c,t} \quad \forall c, t \tag{29}$$

$$aux_g^{\mathrm{sec}}_{c,t} \leq M \left(1 - u^d_{c,t}\right) \quad \forall c, t \tag{30}$$

$$\sum_c \left(\frac{g^{\mathrm{sec,up}}_{c,t}}{g^{\mathrm{sec,up}}_{c,t} + g^{\mathrm{sec,dw}}_{c,t}} \right) = R^{\mathrm{SM}}_t \quad \forall t. \tag{31}$$

The secondary regulation energy in each period t is remunerated according to the net energy, i.e. regulation-up energy minus regulation-down energy, (32)–(34). A positive net regulation energy (ϕ_t=1) is remunerated according to the regulation-up energy price and the PSHP must discharge extra water from the upper reservoir whereas if the net regulation energy is negative (ϕ_t=0), the PSHP must pay the regulation-down energy price, keeping water in the upper reservoir. If ϕ_t=1, $e^{\mathrm{sec,dw}}_t = 0$ by equation (34) and $e^{\mathrm{sec,up}}_t$ equals the right-side of equation (32). Analogously, if ϕ_t=0, $e^{\mathrm{sec,up}}_t = 0$ by equation (33) and $e^{\mathrm{sec,dw}}_t$ equals the right-side of equation (32)

$$e^{\mathrm{sec,up}}_t - e^{\mathrm{sec,dw}}_t = \sum_c \left(\rho^{\mathrm{up}}_t \cdot g^{\mathrm{sec,up}}_{c,t} - \rho^{\mathrm{dw}}_t \cdot g^{\mathrm{sec,dw}}_{c,t} \right) \quad \forall t \tag{32}$$

$$0 \leq e^{\mathrm{sec,up}}_t \leq \phi_t \cdot M \quad \forall t \tag{33}$$

$$0 \leq e^{\mathrm{sec,dw}}_t \leq (1 - \phi_t) \cdot M \quad \forall t. \tag{34}$$

B. Model Formulation for SC-B PSHP

Model formulation of PSHPs operating in hydraulic SC-B is composed by equations (1)–(34) and equation (35). The latter prevents each unit to be operated simultaneously in generating and pumping mode (see Section II for details of binary units). Due to this, the I/O curve of each binary unit is the same as the one of Conv PSHPs, i.e., the one presented in Fig. 2 but with the solid line removed. However, note that the plant can be operated simultaneously in generating and pumping mode if it is equipped by two or more units, thanks to the hydraulic SC

$$u^d_{c,t} + u^p_{c,t} \leq 1 \quad \forall c, t. \tag{35}$$

C. Model Formulation for Conventional PSHP

Model formulation of Conv PSHPs is composed by equations (1)–(35) and equation (36). A Conv PSHP can be operated in generating or pumping modes but not in hydraulic SC mode. Therefore, the plant only regulates power in generating mode, losing the opportunity to participate in the secondary regulation service while it is pumping, regardless of the number of pump-turbine units. The I/O curve of their units is the same as the one presented in Fig. 2 but with the solid line removed. In addition to this, Conv PSHPs usually comprise a single penstock between the upper reservoir and the power station. Consequently, if there is at least one unit operating in pumping mode, no unit can be operated in generating mode, and vice versa, (36). For instance, if the Conv PSHP is composed by three units, i.e. C = 3, and unit c_1 is pumping, i.e., $u^p_{c_1,t} = 1$, the right-side of equation (36) is zero when the constraint is evaluated for $c = c_1$ and, therefore, the rest of the units of the plant have an off state in generating mode, i.e., $u^d_{c_2,t} = u^d_{c_3,t} = 0$, left-side of the equation. Due to (35), when a unit is in pumping mode, it cannot be in generating mode, i.e., $u^d_{c_1,t} = 0$ as $u^p_{c_1,t} = 1$

$$\sum_{c' \neq c} u^d_{c',t} \leq (\mathrm{C} - 1) - (\mathrm{C} - 1) \cdot u^p_{c,t} \quad \forall c, t. \tag{36}$$

IV. Case Study

A. Data

1) Electric Power System Data: The electric power system data correspond to historical hourly values of: 1) spot prices, 2) secondary regulation reserve prices, 3) secondary regulation-up and down energy prices, 4) percentage of real-time use

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TABLE II
TECHNICAL DATA OF EACH UNIT IN THE PROPOSED PSHPS, FLOWS IN m^3/s, POWER IN MW, EFFICIENCIES IN % AND START-UP COSTS IN €

	Conv and SC-B	SC-T
$\overline{q}^d_{c,k}$	115.7/115.7	115.7/115.7
$\overline{g}^d_{c,k}$	297.4/314.1	297.4/314.1
$\overline{\eta}^d_{c,k}$	90/92	90/92
$\underline{q}^d_{c,k}$	49.77/49.77	23.15/23.15
$\underline{g}^d_{c,k}$	107.9/114.5	52.87/55.99
$\underline{\eta}^d_{c,k}$	76/78	80/82
$\overline{q}^p_{c,k}$	115.7/115.7	115.7/115.7
$\overline{g}^p_{c,k}$	389.8/402.8	389.8/402.8
$\eta^p_{c,k}$	90/90	90/90
cSU^d_c	1650.6	1650.6
cSU^p_c	1321.5	1321.5

of regulation-up and down reserves and 5) the ratio between regulation-up reserve and total reserve determined by the TSO, R^{SM}_t.

Actual regulation-up and down energy provided by each power plant in Spain is not publicly available. In order to build the required input data, it is assumed that the hourly real-time use of reserves, (ρ^{up}_t, ρ^{dw}_t) is given by the historical hourly ratio of aggregate power delivery and aggregate assigned reserves in the entire Spanish power system. The maximum hourly real-time use of reserves is assumed to be 100%. In addition to this, hourly regulation reserve ratio, R^{SM}_t is assumed to be the historical hourly ratio of aggregate secondary regulation-up and total reserve requirements of the entire Spanish power system.

2) Technical Data: Technical data of 1) the reversible Francis pump-turbines for the Conv PSHP and the SC-B PSHP and 2) the Pelton turbines and pumps at fixed speed for the SC-T PSHP are presented in Table II. All data is split into pairs (except start-up costs): values on the left (resp. right) of the slash refers to the I/O curve for the lower (resp. higher) reservoir level. Higher gross head is considered to be 310 m while lower gross head is assumed to be 300 m. Hydraulic losses, mainly resulting from friction of the water in the penstock, are considered 3% of the gross head [25], i.e., 9.3 m for the higher gross head and 9 m for the lower gross head. It is assumed that the three analysed PSHPs (Conv, SC-B and SC-T) are formed by two identical units.

Maximum water discharge and pumped water flow have been chosen in order to empty and refill the upper reservoir in a minimum time of 6 h (the upper reservoir capacity is 5 hm^3, $\overline{v} = 5\ \mathrm{hm}^3$ and $\underline{v} = 0\ \mathrm{hm}^3$). In generating mode, the minimum flow and efficiencies at maximum and minimum flows are determined from typical performance curves in [25]. A linear relationship between water discharge and power generation is assumed. As all pumps (and reversible Francis pump-turbines in pumping mode) operate at fixed speed, they have a single operating point with a typical efficiency of 90% [26]. The round-trip efficiency of the ternary units is between 78% at the maximum operating point and 69.5% at the minimum operating point for the higher gross head and between 76.3% and 67.8% for the lower gross head. In addition to this, the round-trip efficiency of the binary units is between 78% at the maximum operating point and 66.1% at the minimum operating point for the higher gross head and between 76.3% and 64.4% for the lower gross head.

Finally, start-up costs of turbines and pumps have been calculated according to [27]. No differences have been considered between reversible Francis in generating mode and Pelton turbines or between reversible Francis in pumping mode and pumps of SC-T PSHPs.

B. Methodology

The three proposed models in Section III have been applied, respectively, to the closed-loop and daily-cycle PSHPs of Table II to calculate the maximum theoretical income in the above-mentioned markets and services in two representative cases. For this purpose, the model was run, day by day, over two different periods of one week (one in summer and one in late autumn) and with a daily initial and final volume in the upper reservoir of 2.5 hm^3, in order to 1) determine the extra value that the hydraulic SC operation introduces comparing to Conv PSHPs participating in the spot and the secondary regulation service and 2) analyse and compare to each other the power and reserve schedule of Conv PSHPs and PSHPs operating in hydraulic SC mode. Each daily problem is solved with a branch and cut algorithm in Cplex in a 2.2 GHz Dual Core AMD Turion CPU, with 4 GB of RAM memory. Hundreds seconds of CPU time were necessary for each week and each PSHP.

C. Maximum Theoretical Income

For Conv, SC-B and SC-T PSHPs, maximum theoretical income in the two cases under study and in the day 11/11/2014, whose operation is discussed in Section IV-D, are shown in Table III. These results consider that all PSHPs are formed by two identical units with the data of Table II. Revenues are composed by the spot market income (DM), secondary regulation reserve market income (SM) and income from secondary regulation-up (ER2UP) energy. Costs are related to secondary regulation-down energy (ER2DW) and start-up costs for turbines, cSU^d and pumps, cSU^p. In addition to this, income in the spot market is divided into sold energy when the plant is generating (DM^+) and purchased energy when is pumping (DM^-).

All PSHPs are mainly obtaining their revenues from the secondary regulation reserve market. As the PSHP is more flexible (SC-T PSHPs are more flexible than SC-B PSHPs that are more flexible than Conv PSHPs), the maximum theoretical income from this market and in total are higher, whereas the spot market revenue is lower and even negative (see DM in the first case for all PSHPs and in the second case for the SC-T PSHP). This reflects that the PSHP is using the secondary regulation reserve as the priority source of revenue, and the spot market as a subsidiary means to maximize the participation in the secondary regulation market.

As a consequence of a higher participation in the secondary regulation reserve market, not only the income from the said market is higher as the PSHP is more flexible, but also the

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TABLE III
MAXIMUM THEORETICAL INCOME OF PSHPS, [10^3 €]

		Conv	SC-B	SC-T
	DM^+	1663.35	1729.06	2526.51
	DM^-	−1721.60	−1798.44	−2924.44
First Case	DM	−58.25	−69.38	−397.93
30/06/2014	SM	707.28	728.43	1682.55
to	ER2UP	131.68	140.07	441.38
06/07/2014	ER2DW	−165.02	−171.92	−287.37
	cSU^d	−66.02	−64.37	−34.66
	cSU^p	−34.36	−33.04	−35.68
	Total	515.31	529.80	1368.28
	DM^+	1409.07	1477.76	2102.60
	DM^-	−1332.60	−1430.89	−2599.97
Second Case	DM	76.47	46.87	−497.36
10/11/2014	SM	1163.93	1180.58	2969.99
to	ER2UP	231.26	250.55	765.18
16/11/2014	ER2DW	−81.84	−82.27	−148.03
	cSU^d	−85.83	−62.72	−33.01
	cSU^p	−58.15	−43.61	−50.22
	Total	1245.84	1289.39	3006.56
	DM^+	248.80	262.76	325.40
	DM^-	−236.18	−257.42	−395.03
	DM	12.62	5.34	−69.63
Day	SM	150.22	162.80	386.76
11/11/2014	ER2UP	31.40	35.40	112.28
	ER2DW	−18.11	−22.48	−35.95
	cSU^d	−9.90	−8.25	−4.95
	cSU^p	−7.93	−6.61	−7.93
	Total	158.29	166.19	380.57

profit from the regulation-up and down energy increases. It is important to note that in the Spanish electric power system, the secondary regulation-up (resp. down) energy prices are usually higher (resp. lower) than the spot market prices (in 2014, the mean spot market price was 42.15€/MW·h whereas the mean secondary regulation-up and down energy prices were 47.36 and 31.34 €/MW·h, respectively). Due to this and also due to the importance of the secondary regulation reserve price, a sensitivity analysis of the maximum theoretical income to a variation in the secondary regulation reserve price and in the secondary regulation-up and down energy prices is presented in Section IV-E.

Comparing maximum theoretical income between PSHPs, the highest increase is between Conv and SC-T PSHP, with 165.5% higher income of SC-T PSHP in the first case, and 141.33% in the second case. The reason for such significant increment is the high flexibility that hydraulic SC operation with ternary units introduces: in all hours in which Conv PSHP is in pumping mode, it cannot participate in the reserve market whereas SC-T PSHP is able to participate with a regulation range equivalent to that of all turbines in operation. Furthermore, when a unit in the Conv PSHP is operating in pumping mode, this unit and the rest must be off in generating mode whereas when a unit in the SC-T PSHP is operating in pumping mode, this unit and the rest can be on or off in generating mode. Increment in maximum theoretical income between Conv and SC-B PSHP is significantly reduced to 2.81% in the first case and 3.5% in the second case. The reason for this small increase is because SC-B PSHP is much less flexible than SC-T PSHP: in the former, the hydraulic SC operation is restricted to the situation where one unit is operated in pumping mode and the other in generating mode.

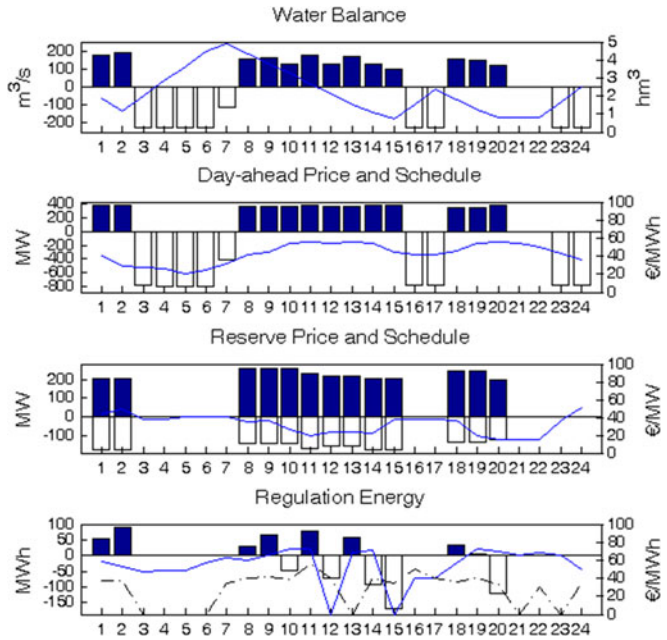


Fig. 3. Optimal operation of Conv PSHP. In regulation energy, regulation-up (resp. down) energy prices are in solid (resp. dashed) line.

D. PSHP Operation in Hydraulic SC Mode

The energy and reserve schedule of the Conv PSHP, and the amount of regulation-up and down energy requested in real-time to the plant, according to the hypotheses described above, are shown in Fig. 3, for 11/11/2014; the same variables are shown in Fig. 4 for the SC-T PSHP. The operation of the SC-B PSHP is similar to the one of the Conv PSHP. Each figure is divided into four subfigures where the following variables are depicted: 1) upper reservoir water volume in solid line and total water through the turbines (positive) or pumps (negative) in bars, 2) energy schedule in bars and price of the spot market in solid line, 3) reserve schedule in bars and price of the reserve market in solid line and 4) regulation-up and down energy in bars and prices in solid and dashed lines, respectively.

The most important difference between the operation of the Conv and SC-T PSHPs comes, as expected, from the fact that the former can only participate in the secondary regulation reserve market in generating mode, whereas the latter can participate in the said market both in consumption and generating modes. As can be seen in the figures, both PSHPs pump/discharge water to/from the upper reservoir during off-peak/peak hours (lower/higher spot market prices).[2] The Conv PSHP manages to make some profit from the difference between peak and off-peak spot market prices (the so-called price arbitrage) (see economic results of 11/11/2014 in Table III), whereas the SC-T PSHP obtains a negative revenue in the spot market.

[2]There are also some hours with intermediate prices where the PSHP pumps or discharges water in an apparently arbitrary manner, in order to reduce the number of start-ups and to respect the reservoir storage limits.

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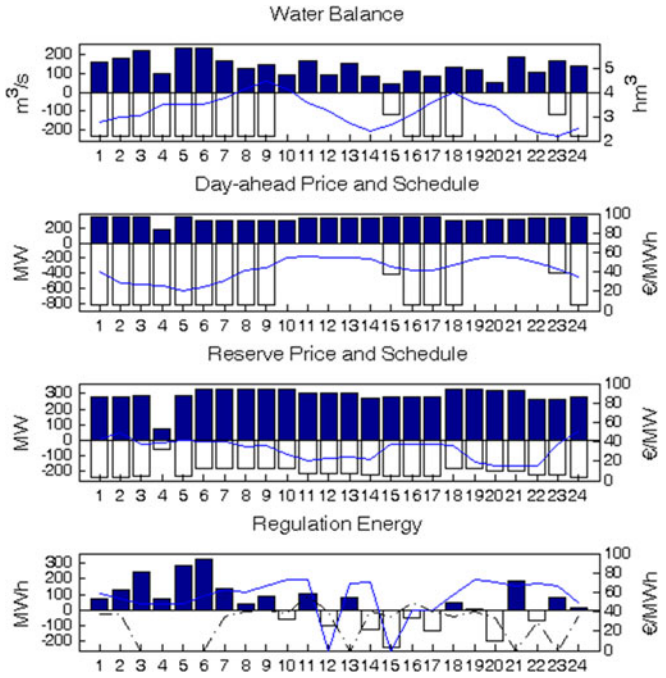


Fig. 4. Optimal operation of PSHP in hydraulic SC with ternary units. In regulation energy, regulation-up (resp. down) energy prices are in solid (resp. dashed) line.

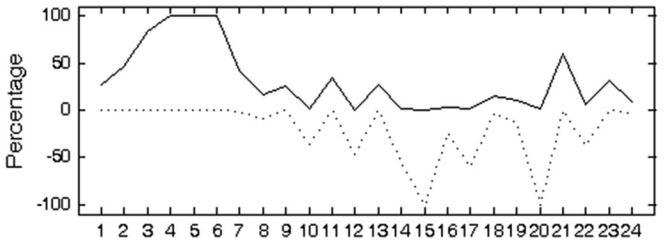


Fig. 5. Real-time use of reserves, regulation-up (resp. down) uses are in solid (resp. dotted) line.

However, the SC-T PSHP benefits also from the reserve prices during off-peak hours in the spot market, which according to unpublished studies carried out by the authors, do have a negative correlation with the spot market prices, and are, therefore, usually higher during off-peak hours of the spot market in the Spanish electricity system. Theoretically, the Conv PSHP could offer energy and reserve in hour 3, in which the reserve price is quite high. However, as can be seen in Fig. 5, the TSO would ask the PSHP to provide around 85% of the offered reserve, which might cause the storage capacity in the upper reservoir to run out. Instead, the Conv PSHP chooses to fill up the upper reservoir from hour 3 to hour 7, and to offer again energy and reserve from hour 8 to hour 15. Of course, the Conv PSHP could choose to alternate one or two hours in pumping mode followed by one or two hours in generating mode, in order to benefit from the higher reserve prices, at the expense of increasing the start-up costs. In Fig. 4, it is even more evident that the operation of the SC-T PSHP is focused on making profit from the secondary regulation reserve market. The SC-T PSHP operates in hydraulic SC mode during peak hours in the reserve market

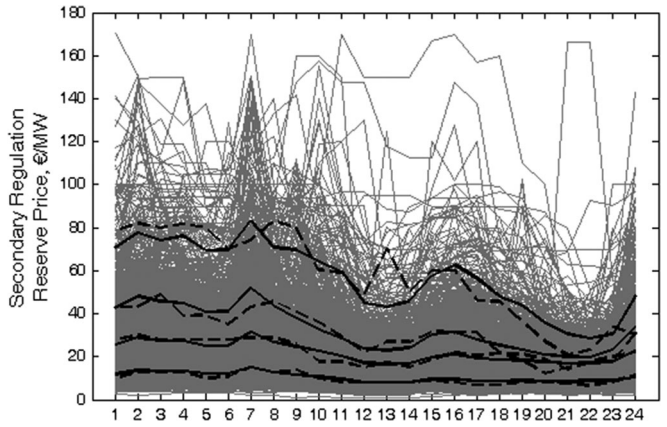


Fig. 6. Clusters (black solid line) and closest historical price profiles (dashed line) for the sensitivity analysis of the secondary regulation reserve price.

and in generating mode during peak hours in the spot market, and offers reserve every single hour.

Finally, it is interesting to note that while the Conv PSHP makes use of the whole storage capacity, the SC-T PSHP uses only a small fraction of its capacity. Even though it is not possible to draw reliable conclusions from the operation of a single day, this could have a positive impact on the storage capacity sizing, and therefore on the investment cost of the PSHP. However, this issue is outside the scope of the paper, and is suggested as a future line of work in Section V.

E. Sensitivity Analysis

Three sensitivity analyses have been carried out in order to study the sensitivity of the maximum theoretical income to a variation in 1) the secondary reserve price, 2) the secondary regulation-up energy price and 3) the secondary regulation-down energy price. In each sensitivity analysis, price profiles are chosen according to the following methodology.

1) The hourly historical data of the price under study in each day of the period 2010 to 10/11/2014 is clustered into four daily groups based on the method of the squared Euclidean distances [28].
2) The real daily historical price profile that is the closest to each centroid is chosen; the distance between each historical price profile and each centroid is measured from the Euclidean distance. Fig. 6 shows the centroids and the closest price profiles of the historical data used in the sensitivity analysis corresponding to the secondary regulation reserve price.
3) The other two price profiles are chosen directly according to the historical data in the days of the previous step. For instance, the prices profiles for the highest centroid in Fig. 6 correspond to 28/12/2013, being the real prices of the secondary reserve and the secondary regulation-up and down energy of this day the ones that are used.
4) The spot prices, the percentages of real-time use of regulation-up and down reserves and the ratios between the regulation-up reserve and the total reserve, R_t^{SM}, are those of the day 11/11/2014.

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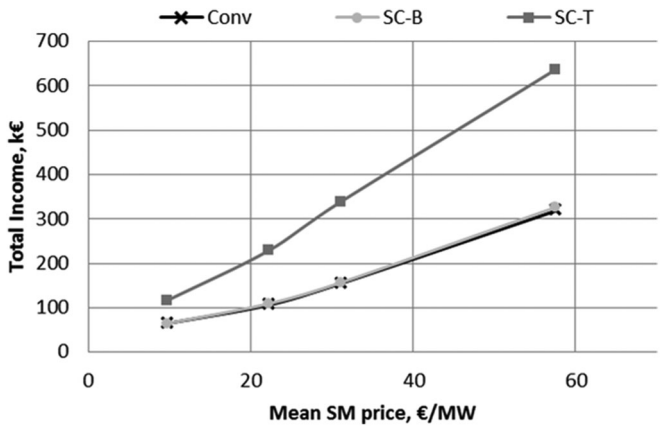


Fig. 7. Sensitivity analysis of the secondary regulation reserve price.

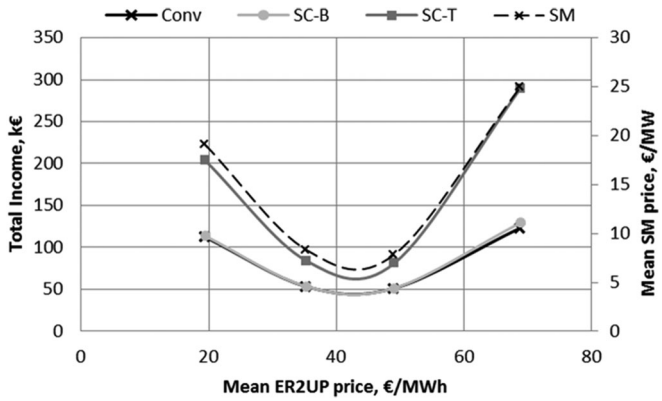


Fig. 8. Sensitivity analysis of the secondary regulation-up energy price.

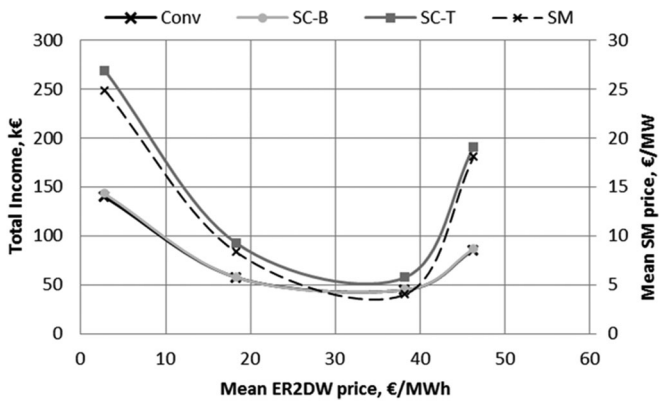


Fig. 9. Sensitivity analysis of the secondary regulation-down energy price.

The results of the sensitivity analyses with respect the secondary regulation reserve price, the secondary regulation-up energy price and the secondary regulation-down energy price are shown in Figs. 7–9, respectively. In each figure, the maximum theoretical income of Conv, SC-B and SC-T PSHPs under the above-described assumptions is presented in the left *y*-axis. In addition, the average value of the price under study in the sensitivity analysis is shown in the *x*-axis. In the sensitivity analyses of the secondary regulation-up and down energy prices, the average value of the secondary regulation reserve price is also shown in the right *y*-axis in order to better understand the behaviour of the maximum theoretical income.

The sensitivity of the maximum theoretical income to the mean secondary regulation reserve price is significant, Fig. 7, varying the former sixfold when the latter varies also sixfold. In the day under study, the sensitivity seems to be quite linear, being higher for SC-T PSHP than for Conv and SC-B PSHPs, whose sensitivity is almost the same. This sensitivity analysis confirms the results obtained in Section IV-C because the income from the secondary regulation reserve market are the most significant and, therefore, a change in its mean price affects considerably the total maximum theoretical income.

The sensitivity of the maximum theoretical income to the mean secondary regulation-up energy price is also significant, Fig. 8, and also higher for SC-T PSHP than for Conv and SC-B PSHPs, whose sensitivity is again almost the same. However, the total income curves are again explained by the mean SM price, first decreasing and then increasing as the mean secondary regulation-up energy price increases. This result also confirms the ones obtained in Section IV-C regarding the small magnitude of the income from the secondary regulation-up energy comparing to the total maximum theoretical income.

Finally, the sensitivity analysis of the mean secondary regulation-down energy price is similar to the one with the mean secondary regulation-up energy price, Fig. 9.

V. Conclusion

An MILP model for the energy and reserve scheduling of a PSHP considering hydraulic SC operation is presented. Additionally, this study compares maximum theoretical income of a conventional PSHP and a PSHP operating in hydraulic SC mode with binary and ternary units, in the spot and secondary regulation reserve markets of the Spanish electricity system. The plant under study is a closed-loop, daily-cycle and price-taker PSHP. The results obtained in the article show that a PSHP operating in hydraulic SC mode could enlarge its maximum theoretical income up to 162.3% comparing to a conventional PSHP. In addition to this, the secondary regulation reserve market proves to be the main revenue source for the PSHPs analysed in this article, whereas the spot market would assume a subsidiary role. It is important to consider these results in the context of a price-taker PSHP and a bidding procedure which assumes perfect knowledge in all the data of the electricity system. Further research, taking into account stochasticity in hourly prices and real-time use of reserves, is required to obtain more realistic figures on the expected revenue of a PSHP operating in hydraulic SC mode. Furthermore, considering a price-maker approach in the secondary regulation market is deemed as another future work, where the quota of each participant may be of considerable size. Other trading options of the PSHP in, for example, intraday markets and the tertiary regulation service are also part of future work. Finally, some results obtained in the article suggest the need of further research on the optimal sizing of a PSHP operating in hydraulic SC mode.

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